

OpenMP Lite Offloading Support for Tightly Coupled Accelerators

NXP-RO | ISOLDE | 4.2.10 | Version 0.1

Schema Version			0.9
Basic Information			
IP Name		OpenMP Lite Offloading Support for Tightly Coupled Accelerators	
Version		0.1	
Source Languages		• C • C++	
TRL		TRL-4	
Status		Prototype implemented	
Comment		OpenMP target offloading compiler support for heterogeneous platforms with tightly coupled accelerators, enabling host-controlled accelerator execution and task dispatch	
Provider	Item 1 / Organization Name	NXP Romania	
Provider	Item 1 / Organization URL	www.nxp.com	
Provider	Item 1 / Organization Alias	• NXP-RO	
License		Proprietary	
Repository URL		www.nxp.com	
Project		ISOLDE	
Work Item		4.2.10	
IP Type	Category	• Compiler • Software Library/Framework	
IP Type	Domain	• Other	
IP Type	Comment	OpenMP compiler extensions targeting RISC-V asynchronous TCA	
IP Type	Service	no	
System-Level Information			
Architecture		RISC-V (RV32)	
Processor		Ibex RV32IM	
ISAExtension		RV32IMZicsrXaia	
SW Dependencies		LLVM/Clang 19	
Requires Specialized Tool		No	
Interfaces and Artifacts			
CLI		• Standard clang command-line interface	
API		• Support for a subset of OpenMP 5.x target directives and the taskwait directive	
Input Artifacts	Item 1 / Artifact	C/C++ source file containing OpenMP target pragmas	
Input Artifacts	Item 1 / Format	.c / .cpp	

Output Artifacts	Item 1 / Artifact	Host executable with accelerator activation or embedded image
Output Artifacts	Item 1 / Format	ELF
Output Artifacts	Item 2 / Artifact	TCA Offload Code Generation Report
Output Artifacts	Item 2 / Format	text
Deployment		
Bundle		binary